

Caio Valerio Goulart Correia

Backend Engineer (TypeScript/Node.js) · APIs & Developer Tools · AI Specialist

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Backend engineer with 9+ years of technical experience building and maintaining APIs, services, and developer-facing tools — primarily in TypeScript and Node.js, with deep Python alongside. Shipped and operates a production HR/operations platform (200+ employees) whose Node.js/TypeScript backend exposes 600+ API routes secured by JWT authentication, two-factor authentication, and role-based authorization (PostgreSQL Row-Level Security + ACL) under strict LGPD/PII constraints. Defines cross-product API contracts (HMAC-authenticated sync between systems) and hardens third-party API integrations with retries and fallback on rate-limit/503 responses. Maintains open-source developer tools on GitHub, including MCP tool servers and an SGLang inference-server manager. AI depth: LLM training and evaluation (SFT, DPO/RLHF, LoRA/QLoRA), local serving (vLLM, SGLang, llama.cpp), and RAG. PostgreSQL/Supabase, Docker, GitHub Actions CI/CD, AWS. Fluent English (C1), remote-first experience with international teams. Mechanical engineering background.

PROFESSIONAL EXPERIENCE

AI Specialist (Contract)

micro1 • Aug 2026 – Sep 2026 • Remote

- Trained and evaluated LLMs for frontier-model programs, running Supervised Fine-Tuning (SFT) and preference optimization (DPO/RLHF) with PyTorch, Hugging Face PEFT (LoRA/QLoRA), Unsloth, and Axolotl.
- Ran adversarial testing (red teaming) to surface hallucinations, logic flaws, and bias, feeding corrections back into training datasets.
- Reviewed and benchmarked AI-generated code across Python, TypeScript, Node.js, FastAPI, React, Next.js, and JavaFX — assessing data-structure choices, execution behavior, and complex API integrations.
- Deployed local LLM serving and agent harnesses on Windows/Linux (vLLM, SGLang, llama.cpp) on dedicated NVIDIA hardware (L4 GPU, 24GB VRAM), plus MCP tool servers for retrieval and graph tools; measured latency/throughput with custom evaluation harnesses.
- Curated high-quality training and synthetic datasets (Self-Instruct, Evol-Instruct) with automated validation for dataset integrity, schema compliance, and model alignment.

Full-Stack Developer & DevOps Engineer ABZ Serviços • Mar 2025 – Present • Rio das Ostras, RJ

- Architected and shipped EmployeeHub (portal.groupabz.com), an HR/operations portal used by 200+ employees — personnel records, crew management, e-Social, vacation, reimbursement, and ACL — on Next.js 15, TypeScript, and Supabase (PostgreSQL with Row-Level Security) on Vercel; reduced HR manual workload by ~40% and approval time by ~95%.
- Built and maintains the platform's Node.js/TypeScript backend — 600+ REST API routes (Next.js App Router) with JWT authentication, two-factor authentication (2FA), and role-based authorization via PostgreSQL Row-Level Security and ACL, protecting sensitive HR/PII data under LGPD.
- Delivered the ABZ product suite around the portal: PontoFlow (bilingual offshore timesheets with manager approval, field annotations, and deadline locks enforced in UI and RLS), Invoice ABZ (FastAPI/SQLAlchemy engine turning payroll and charge-rate spreadsheets into client invoices), and Ticket-Manager (support automation via Microsoft Graph).
- Defined the cross-product API contract (HMAC-authenticated import/export) between the portal and the PontoFlow timesheet product, keeping collaborator identity and period behavior consistent across both systems.
- Led the cloud migration of legacy on-premise Windows Servers (2012–2025), reaching 80% completion with zero production downtime.
- Implemented end-to-end CI/CD with GitHub Actions and Docker, administered Proxmox virtualization and AWS S3, and deployed a WireGuard VPN for 50+ remote employees — sharply reducing connectivity support tickets.

Python Developer & Spatial Data Engineer BRTech3D • Mar 2024 – Mar 2025 • Macaé, RJ

- Built high-performance Python pipelines (NumPy, SciPy, Open3D) automating geospatial data processing for dense point clouds (100M+ points), increasing processing speed by 40% and improving data compression across dozens of projects.
- Modeled and optimized PostgreSQL databases with the PostGIS extension for complex spatial queries, designing and maintaining the REST APIs that exposed them to internal processing pipelines.

Topographical Survey Operator III (Laser Scanning) Laser Master Engineering Services • Aug 2023 – Mar 2024

- Operated high-precision 3D laser scanning equipment (Leica) in complex industrial and archaeological environments.
- Led field campaign planning and technical documentation, ensuring millimeter-level precision and data backup redundancy for large-scale data collection.

Topographical Survey Operator II BRTech3D • Dec 2022 – Aug 2023

- Executed field data collection using Leica TLS equipment and introduced Quality Assurance (QA) processes for outlier cleaning and multiple scan registration.

Technical Documentation & Support Specialist SmartStore / Jackkk Connection • Mar 2021 – Dec 2022

- Produced 50+ bilingual technical manuals and system specifications for international clients.
- Provided advanced pre-sales support and detailed hardware/software architecture mapping for complex business proposals.

Training Specialist & Automation Technician Jackkk Connection Comercial • May 2017 – Mar 2021

- Coordinated and delivered technical training for over 500 professionals in PLCs, hydraulics, and industrial automation.
- Developed educational simulations and managed B2B technical sales, building direct relationships with international suppliers.

Technical Team Lead & Specialized Support Diesel Line Cambuí • Aug 2016 – May 2017

- Led specialized field teams in the maintenance of large-scale engines and turbines (CAT, Cummins, MAK).
- Managed English-language technical support tickets for international clients and formally documented failure analysis processes.

Apprentice Technician Diesel Line Cambuí • Oct 2015 – Aug 2016

- Began career providing support to commercial processes, CRM management, and strict application of compliance procedures for diesel and gas systems maintenance.

RELEVANT PROJECTS

CloudSec & FinOps Compliance Auditor Portfolio Product • 2026

- **Live:** cloudsec-finops-auditor.vercel.app · **Repo:** github.com/Caiolino00/cloudsec-finops-auditor
- Production-oriented compliance auditor that evaluates cloud architecture scenarios against CIS / SOC 2 / FinOps-style policies, returning structured risk, cost-impact, citations, and remediation steps.
- Stack: Next.js (App Router) + TypeScript + Tailwind on Vercel; hybrid RAG (BM25 + TF-IDF + Reciprocal Rank Fusion) over versioned policies/*.md; structured Gemini audits (gemini-3.8-flash → gemini-3.6-flash fallback with automatic retry on 503/rate-limit responses); bilingual EN/PT UI; Vitest unit tests and CI evaluation pipeline.

Portal ABZ (EmployeeHub) ABZ Serviços · Internal HR / Operations Portal · Live

- **Live:** portal.groupabz.com · **Repo:** github.com/Caiolino00/EmployeeHub
- Built the ABZ Group corporate portal (EmployeeHub / Painel ABZ) covering personnel records (DP), crew management, Man Schedule, e-Social, vacation, reimbursement, and ACL — the operational source of truth for people and offshore schedules.
- Stack: Next.js 15 + TypeScript + Supabase (PostgreSQL/RLS) on Vercel; payroll derived from portal embarkation data with WK Radar sync, printable payslips, and audited, reversible schedule edits.
- Integrates with the ABZ suite: identity/collaborator data feeds Timesheet (PontoFlow) sync paths; payroll/charge-rate outputs feed Invoice generation for clients.

PontoFlow (Timesheet Manager) ABZ Group · Timesheet + mobile · Standalone (EmployeeHub-ready)

- **Live:** ponto-flow.vercel.app · **Repo:** github.com/Caiolino000/PontoFlow
- Delivered PontoFlow, a bilingual (pt-BR/en-GB) timesheet system for offshore crew with manager approval, field-level annotations, and monthly deadline lock enforced in UI and RLS.
- Stack: Next.js 15 + TypeScript (Node.js server runtime) + Supabase + Nodemailer; email/cron reminders (T-7...T); mobile companion; multi-tenant admin (invites, per-tenant SMTP).
- Built standalone with EmployeeHub-ready sync (HMAC-authenticated import/export) so collaborator identity and timesheet periods can align with the Portal ABZ suite.

Invoice ABZ (Payroll → Invoice) ABZ Group / ABZ International · Finance automation (private repo)

- Built a multi-company Invoice engine that turns payroll/charge-rate spreadsheets into client invoices (FMS, OMEGA, Jifmar vessels) with automatic client detection and parser routing.
- Stack: FastAPI + SQLAlchemy + SQLite + OpenPyXL; preview → invoice number/date → per-person invoices or ZIP; isolated dev/prod databases.
- Closes the ABZ financial loop: Portal/folha and charge-rate data become standardized invoices for ABZ Group and ABZ International without rewriting code per client.

Developer Tools — MCP Servers & SGLang Commander (Open Source) Personal / Local AI stack

- **Repo:** github.com/Caiolino000/sglang-commander
- Built MCP tool servers over a local knowledge graph (Python + SQLite/FTS indexing) so agents retrieve structured neighborhoods instead of stuffing raw context into prompts.
- Wired tool-calling loops to OpenAI-compatible local serving (vLLM / SGLang) and llama.cpp for low-latency/offline paths; custom harnesses for multi-step agents, retries, and tool-success eval. Open source — includes SGLang Commander (repo link), a desktop + web GUI built for developers managing SGLang inference servers.

TECHNICAL SKILLS

AI & LLM Engineering

PyTorch, Hugging Face (Transformers, PEFT, TRL), LoRA/QLoRA, Unsloth, Axolotl, DeepSpeed, SFT, DPO, RLHF, LLM evaluation & red teaming, synthetic data generation (Self-Instruct, Evol-Instruct), RAG, LangChain, LlamaIndex, vLLM, SGLang, llama.cpp, NVIDIA Triton Inference Server, Model Context Protocol (MCP), agent harnesses / tool calling.

Backend

TypeScript & Node.js (Next.js server runtime, REST API routes, Express, cron workers), Python (FastAPI, SQLAlchemy, NumPy, SciPy, Open3D), API design & cross-product API contracts, authentication & authorization (JWT, 2FA, RLS, ACL, HMAC), resilience patterns (retries, fallback, rate-limit/503 handling), GraphQL, Java (Spring Boot), Go (automation scripting).

Frontend

Next.js 15 (App Router), React, Tailwind CSS.

Data & Databases

PostgreSQL, PostGIS, Supabase (Row-Level Security), SQLite (FTS5), NoSQL, ETL/data pipelines, data lakes, geospatial & point-cloud processing.

Cloud & DevOps

AWS (S3), Docker, Kubernetes, Terraform, GitHub Actions (CI/CD), Vercel, Cloudflare, Proxmox, WireGuard VPN, Linux (Ubuntu/CentOS), Windows Server, PowerShell, Microsoft Graph API.

Testing & QA

Vitest, CI evaluation pipelines, automated dataset validation, schema compliance checks, code review.

EDUCATION

B.S. in Mechanical Engineering (in progress) UniFatecie • May 2024 – May 2027 (expected)

Technical Diploma in Industrial Automation ETP Escola Técnica Brasil • Jan 2014 – Jan 2016

LANGUAGES

Portuguese:

Native |

English:

Fluent (C1 — technical and conversational) |

Spanish / Italian:

Professional working proficiency, advanced technical reading.